



Machine Learning

CMPE 257

MidTerm Exam Prep

Instructor: Zara Hajihashemi, Ph.D.

Mid Term Exam Information

Date: Friday, October 2nd, 2026

Location: ENG 337

Duration: 120 minutes

Points: 25%

Mid Term Exam

Format

- Multi-choice questions
- Short Answers
- Computational Questions

Rules

- No laptop or Electronic devices
- Simple Calculator is needed
- No Brainstorming with LLMs
- Mid term is 25% of Final Grade
- One page cheatsheet (front & back)

Question 1: Short Questions (True/False)

- If we train a Naive Bayes classifier using infinite training data that satisfies all of its modeling assumptions (e.g., conditional independence), then it will achieve zero training error over these training examples.

Question 1: Short Questions (True/False)

- If we train a Naive Bayes classifier using infinite training data that satisfies all of its modeling assumptions (e.g., conditional independence), then it will achieve zero training error over these training examples.

False. Even with infinite data and correct Naive Bayes assumptions, the classes may overlap probabilistically, so some training examples can still be misclassified. Correct model assumptions do **not** imply zero classification error.

Question 1: Short Questions (True/False)

- Logistic regression can only be used for binary classification problems.

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- Logistic regression can only be used for binary classification problems.

False. Logistic regression can be extended to multi-class classification, for example using **one-vs-rest** or **multinomial/softmax logistic regression**.

Question 1: Short Questions (True/False)

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True. SVM seeks a decision boundary that **maximizes the margin**, i.e., the distance between the boundary and the closest training points (the support vectors).

Question 1: Short Questions (True/False)

- Logistic regression assumes a linear relationship between the input features and the output variable.

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- Logistic regression assumes a linear relationship between the input features and the output variable.

False. Logistic regression does **not** assume a linear relationship between the input features and the output (y). It assumes a linear relationship between the features and the **log-odds** of the outcome:

$$\log \left(\frac{p}{1 - p} \right) = \theta^T x$$

logistic regression first defines the probability $p = P(y = 1 \mid x)$

and then considers the **odds** $\text{odds} = \frac{p}{1 - p}$

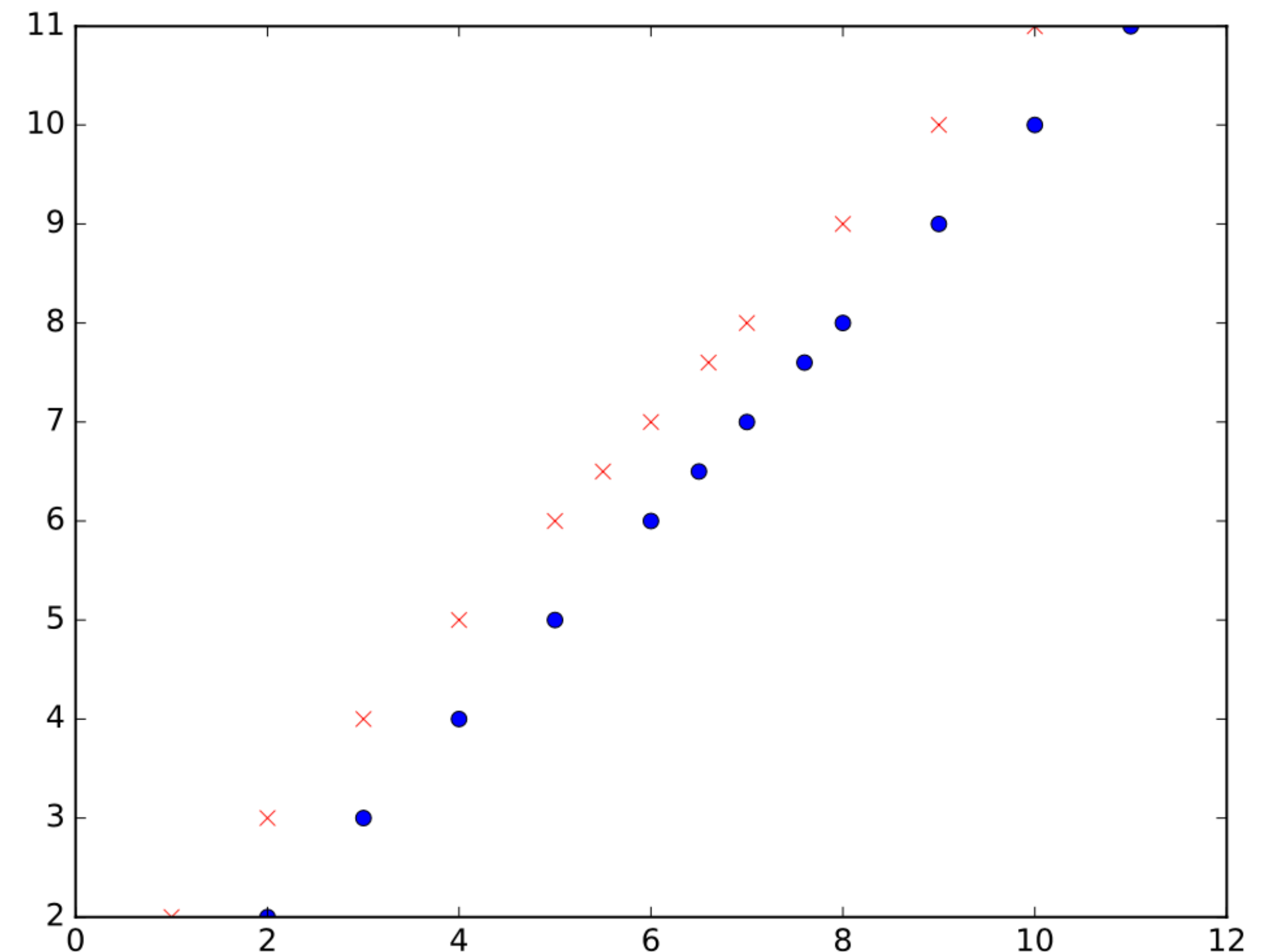
Question 2

- For the given data plots below, choose a method you could use to classify the data, and a method that is not reasonable to use for the given dataset.

Draw (an approximation to) the regions the classifier would classify as positive and negative and briefly explain the performance of two methods per graph.

(a) A supervised learning method that would likely work:

(b) A supervised learning method that would likely not work:



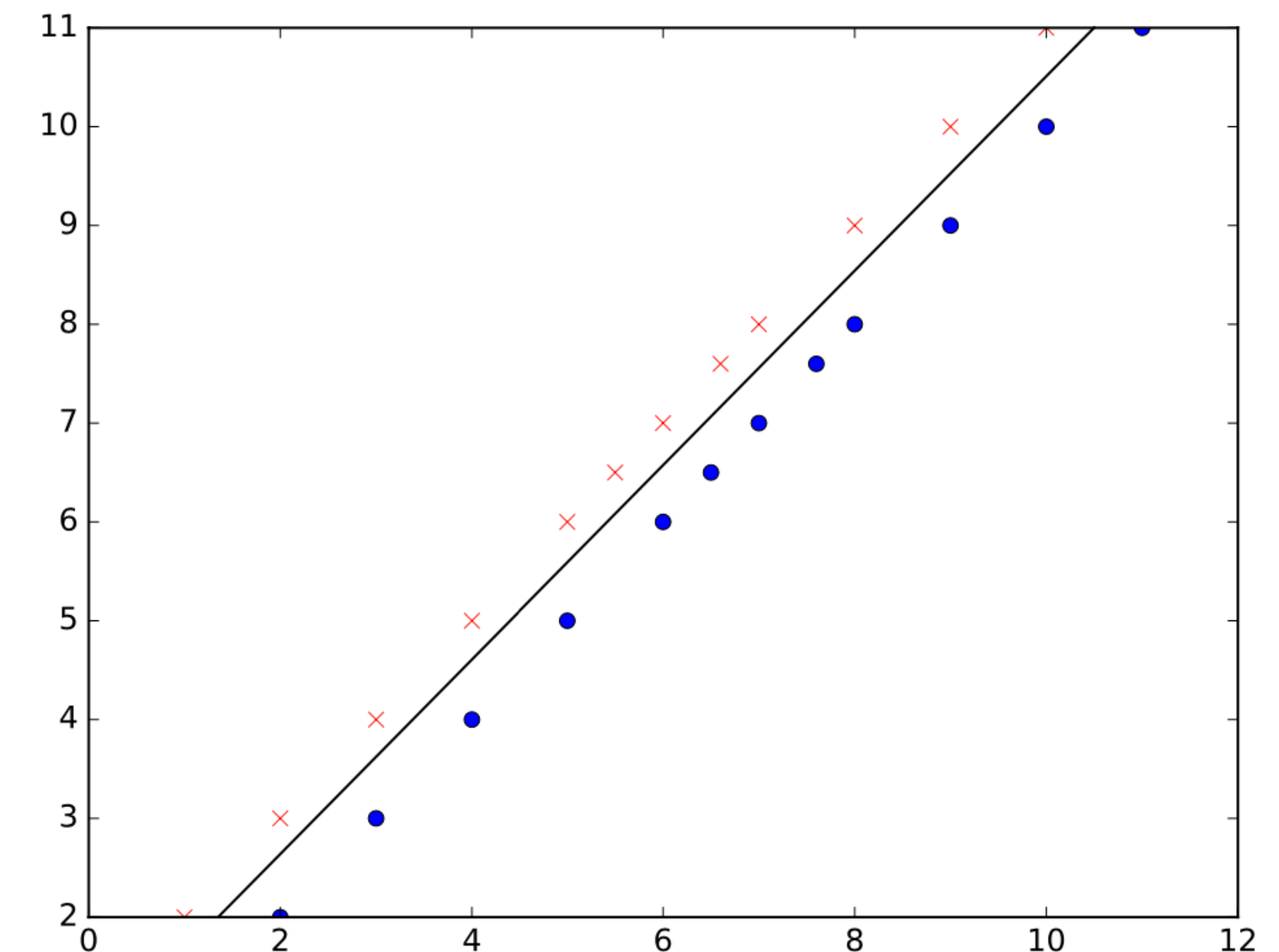
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(a) A supervised learning method that would likely work:

Support Vector Machine - The data points are linearly separable along a diagonal



Question 2

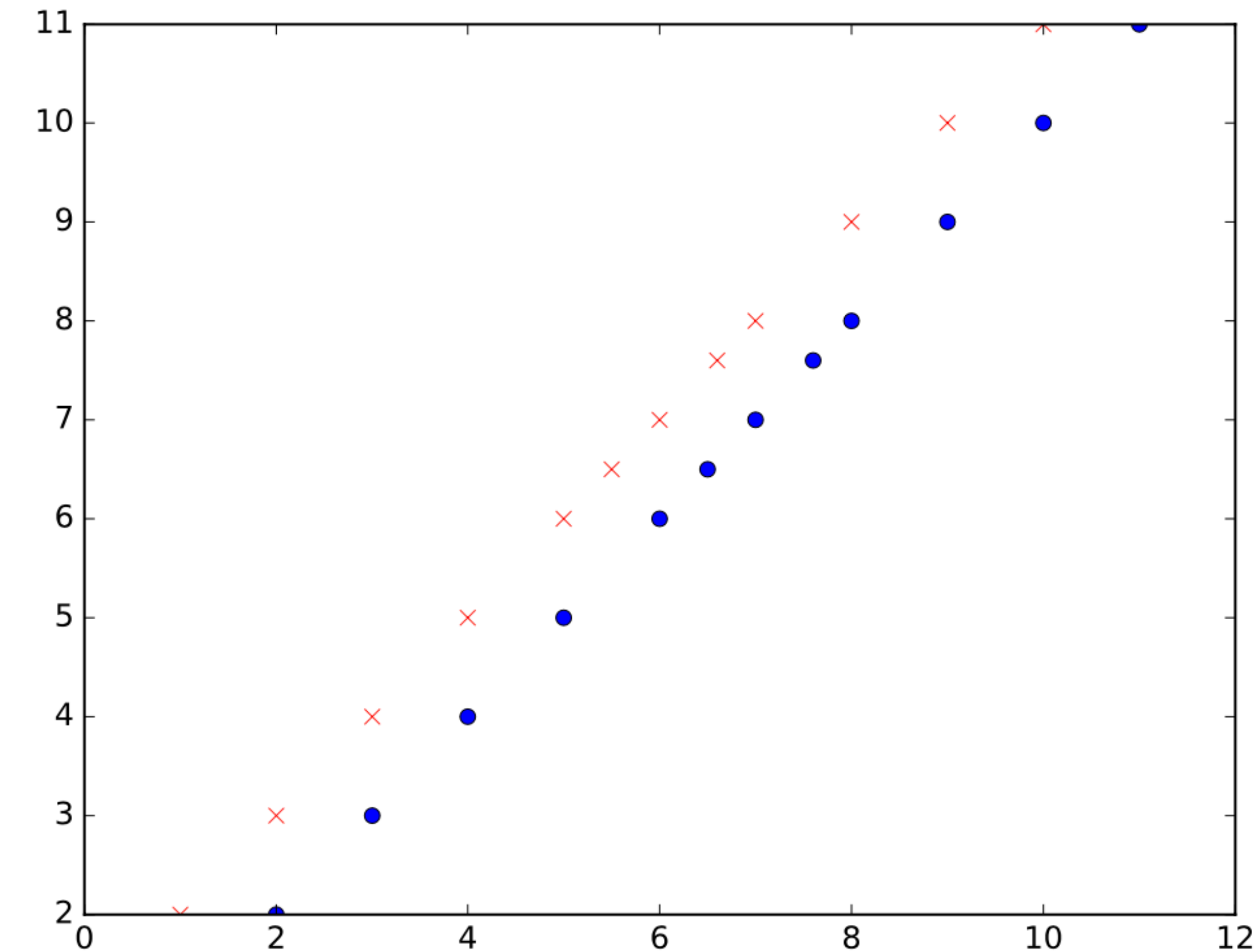
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(b) A supervised learning method that would likely not work:

k-Nearest Neighbors with small k:

Because of sparsity in some regions, KNN may misclassify boundary points.



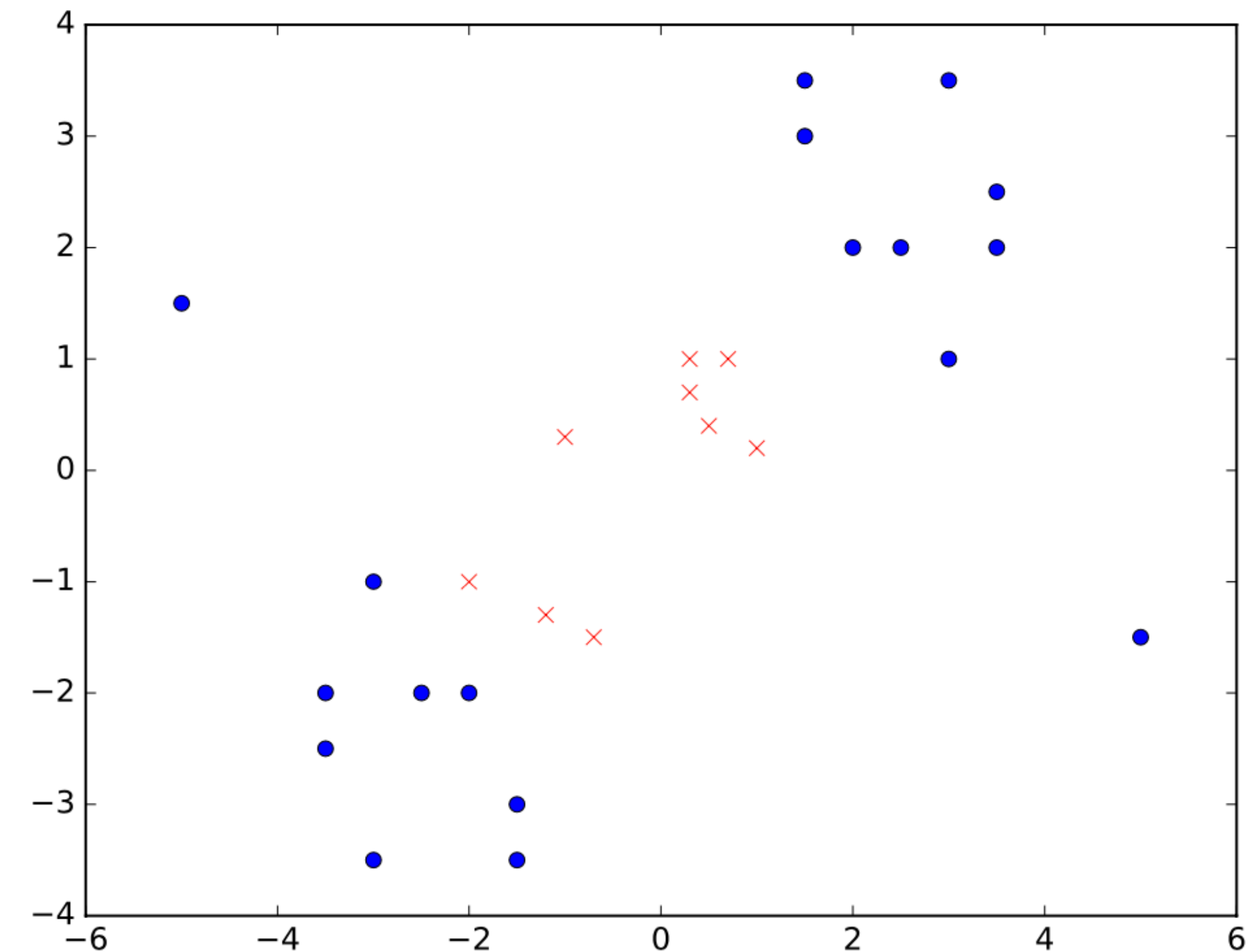
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Question 3

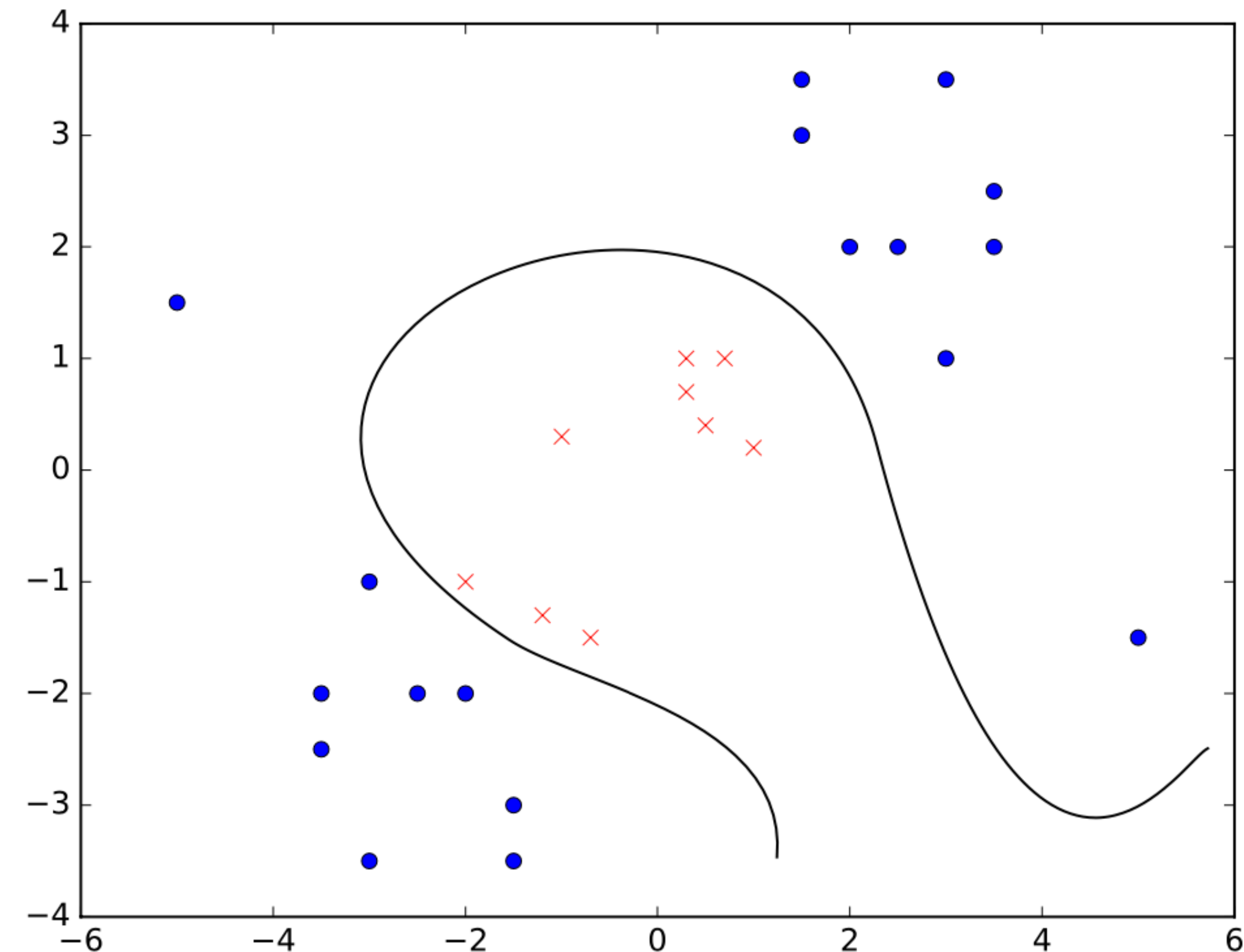
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(a) A supervised learning method that would likely work:

Kernel SVM with polynomial kernel

or logistic regression with polynomial features → Non-linear boundary (a circle) is required.



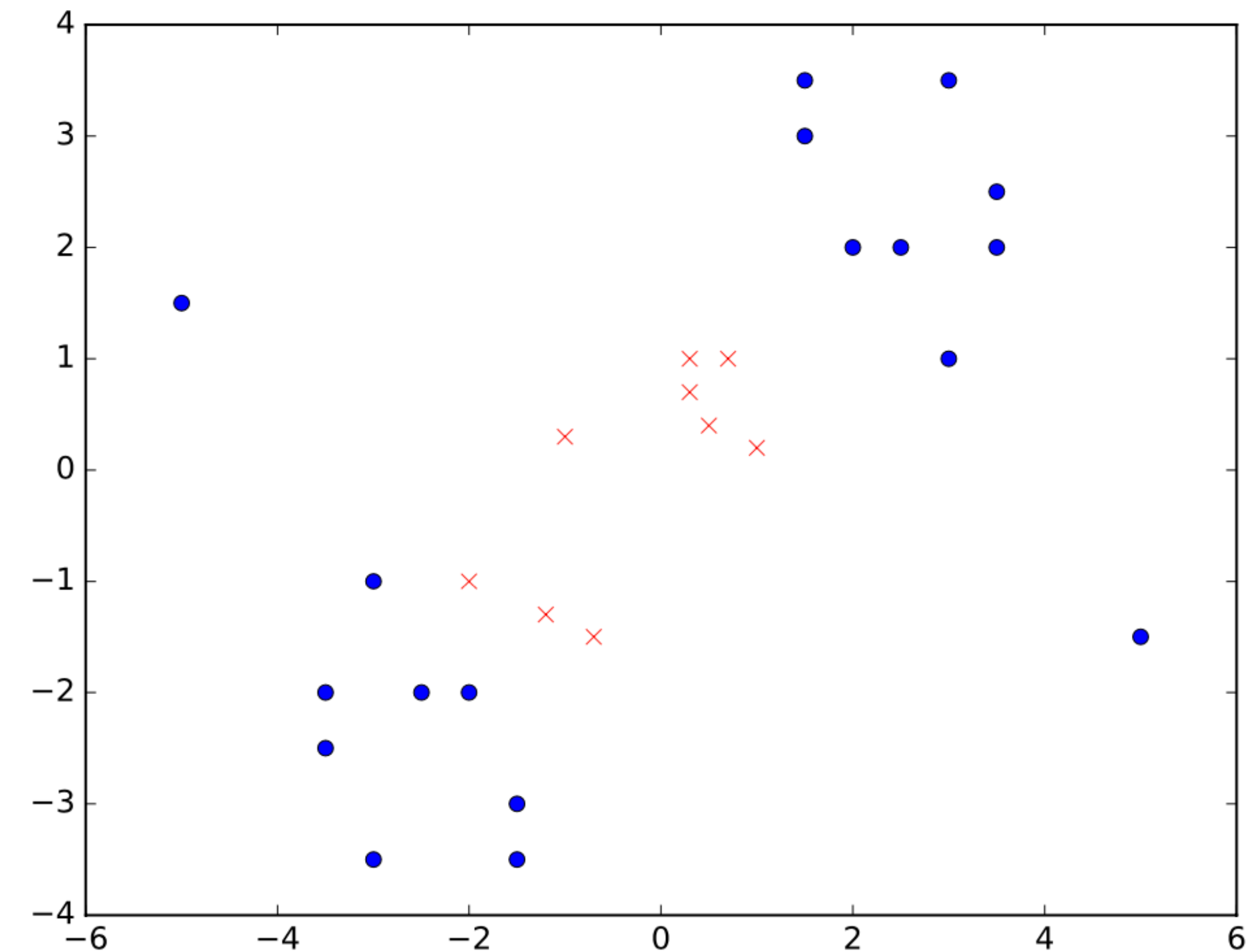
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(b) A supervised learning method that would likely not work:

Linear classification - Data is not linearly separable.



Question 4

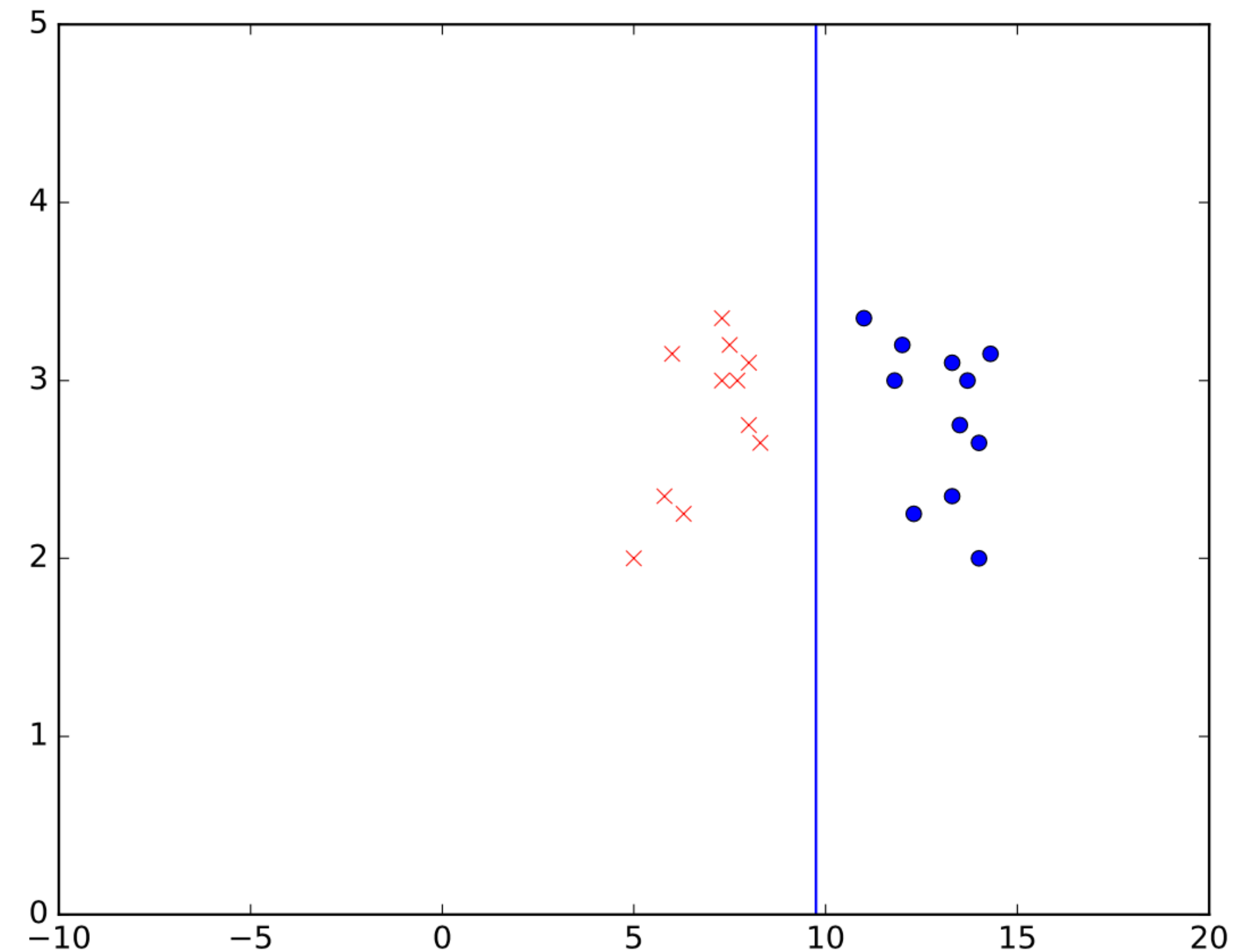
- Let's assume we want to separate the dataset in this figure using the loss functions

$$L_1(\theta^T x, y) = \frac{1}{2}(\theta^T x - y)^2 \quad y \in \{-1, +1\}$$

$$L_2(\theta^T x, y) = [1 - y\theta^T x]_+ = \max\{0, 1 - y\theta^T x\}$$

$$J_1(\theta) = \frac{1}{m} \sum_{i=1}^m L_1(\theta^T x, y)$$

$$J_2(\theta) = \frac{1}{m} \sum_{i=1}^m L_2(\theta^T x, y)$$



1. If We add new data point with positive label at the point (-5, 1), Could the classifying line change for either of the loss functions?
2. Briefly explain why.
3. Draw (your best estimate of) the new classification boundaries, and clearly label the lines with the corresponding loss functions.

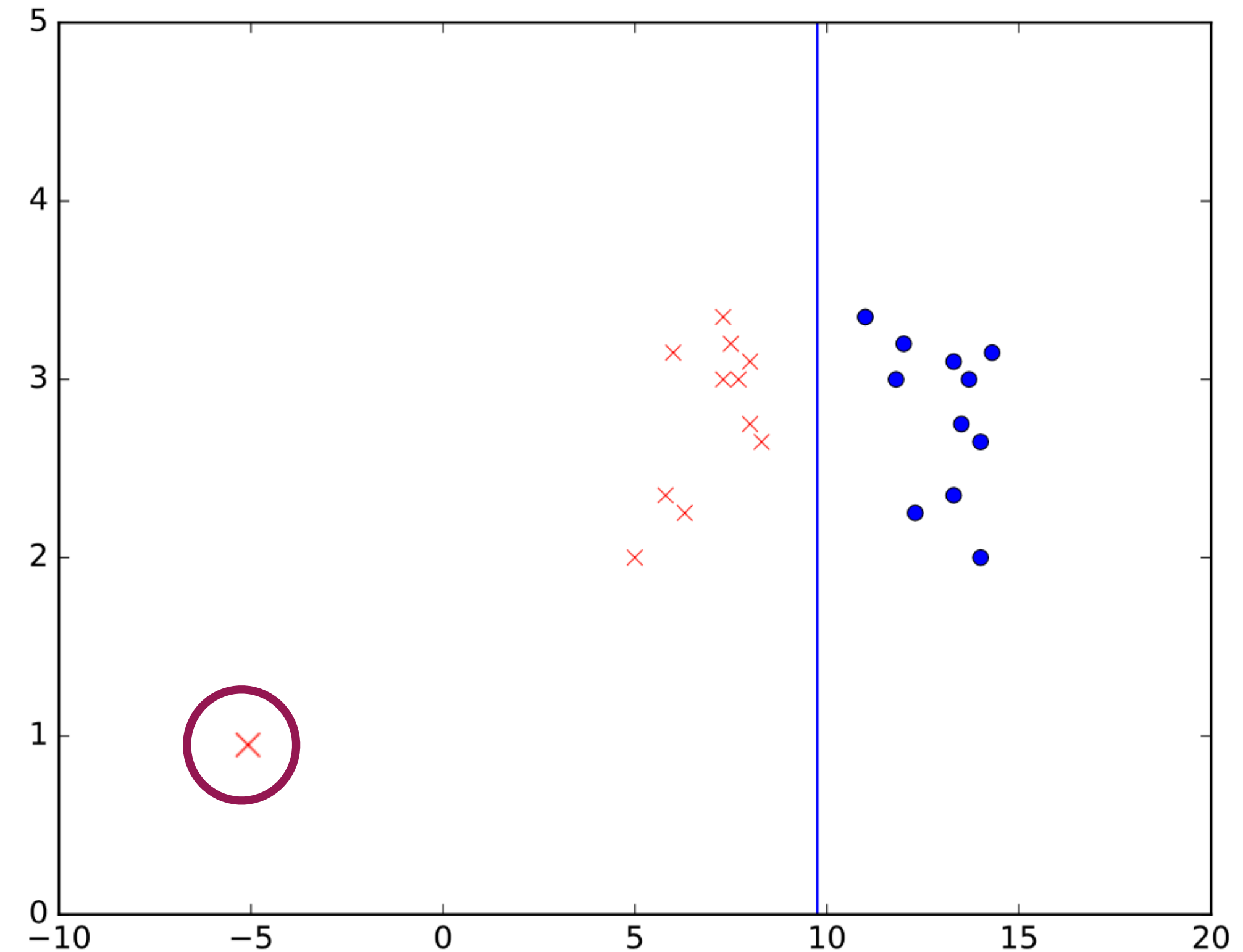
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$$y \in \{-1, +1\}$$



Assume blue points are the positive class $y=+1$ and red crosses are $y=-1$.

Then the new point $(-5,1)$ with a positive label is an **outlier**:

it lies far on the negative/red side of the current decision boundary.

The two losses are:

$$L_1 = \frac{1}{2}(\theta^T x - y)^2$$

$$L_2 = \max(0, 1 - y\theta^T x) \text{ the } \mathbf{hinge \text{ loss}} \text{ used by SVM}$$

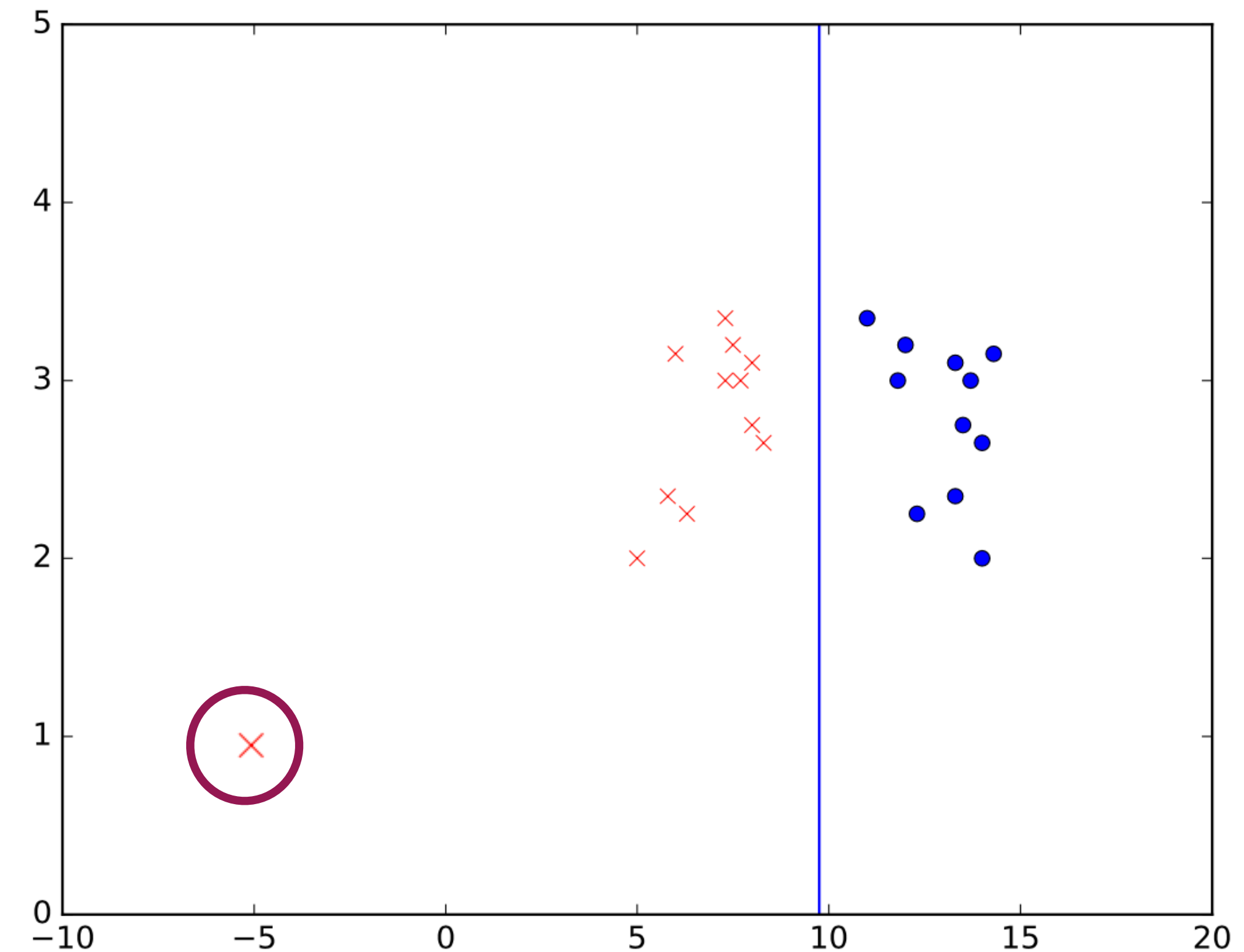
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1. **Yes, for both loss functions.**

But the squared-loss classifier is generally affected much more strongly by this outlier.

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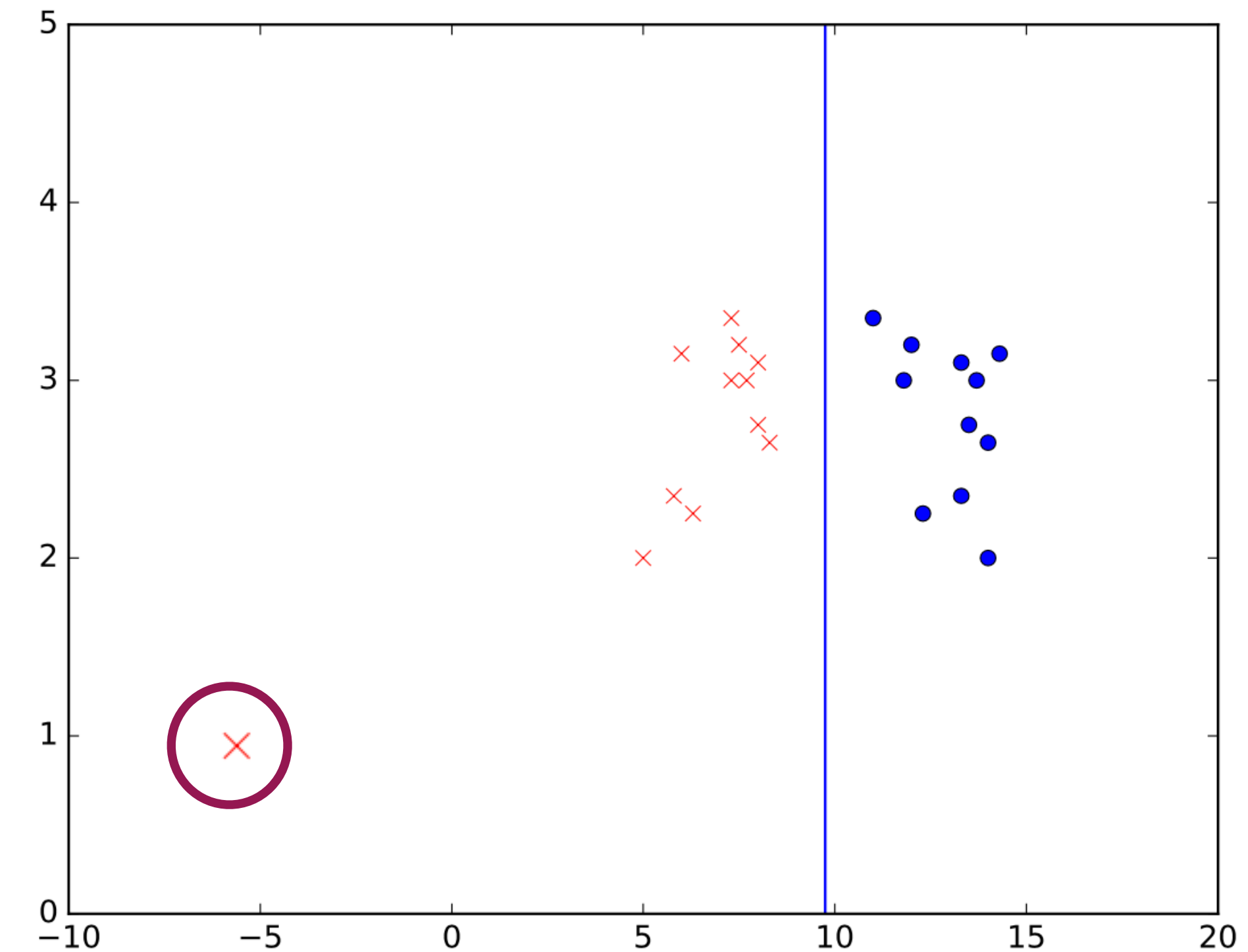
$$L_2(\theta^T x, y) = [1 - y\theta^T x]_+ = \max\{0, 1 - y\theta^T x\}$$

2. Why?

For **L₁: squared loss**, the new point is very far from where a positive example is expected. Its error is squared: $L_1 = \frac{1}{2}(\theta^T x - y)^2$

So if its prediction is very wrong, the penalty becomes very large. For example, an error of 4 gives a loss proportional to $4^2=16$.

Therefore, this one far-away point can strongly pull the fitted classifier toward itself. The decision boundary could **move or rotate significantly**.



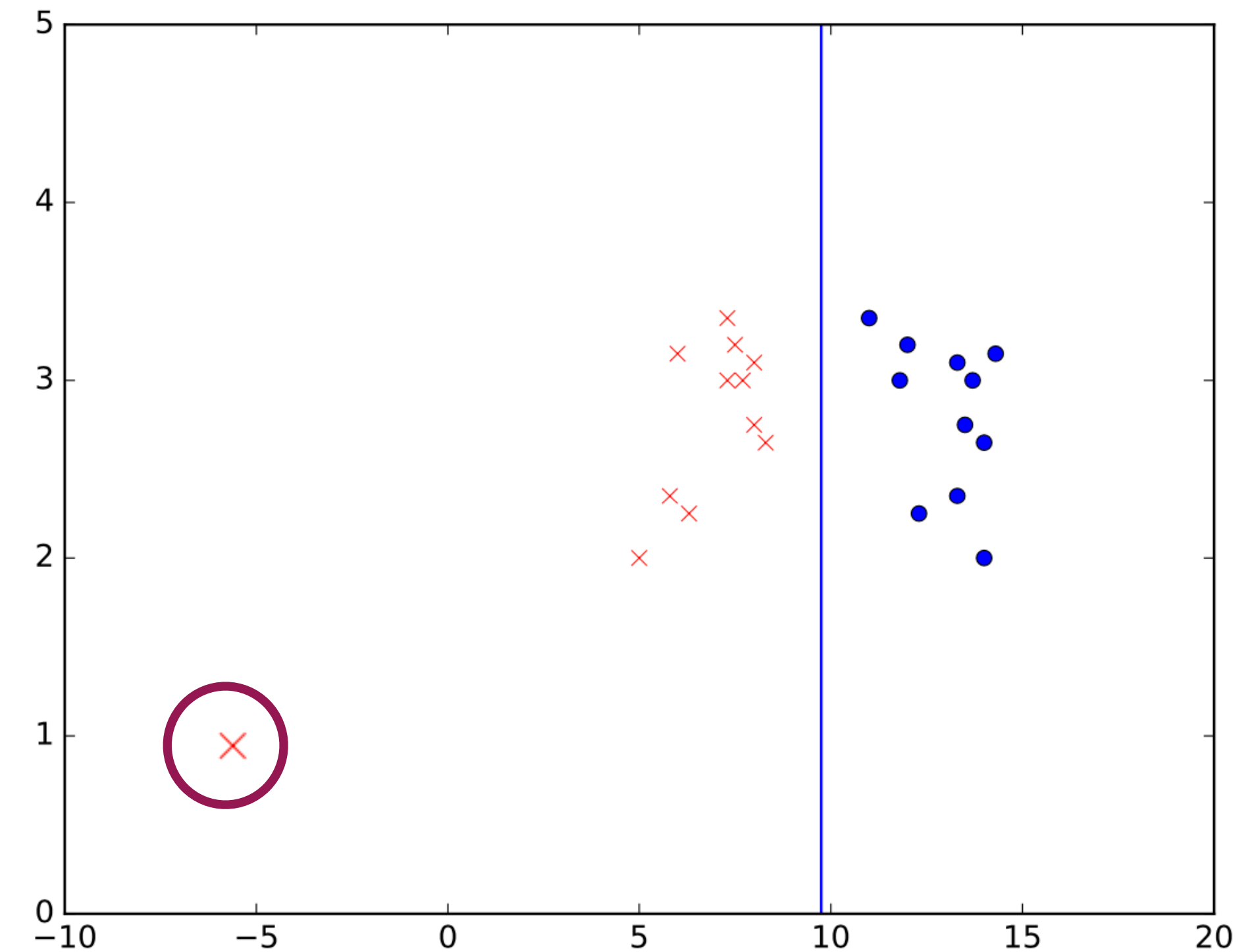
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2. Why?



For L_2 : **hinge loss**, the new point is also misclassified, so it does contribute to the loss. But hinge loss grows **linearly**, rather than quadratically.

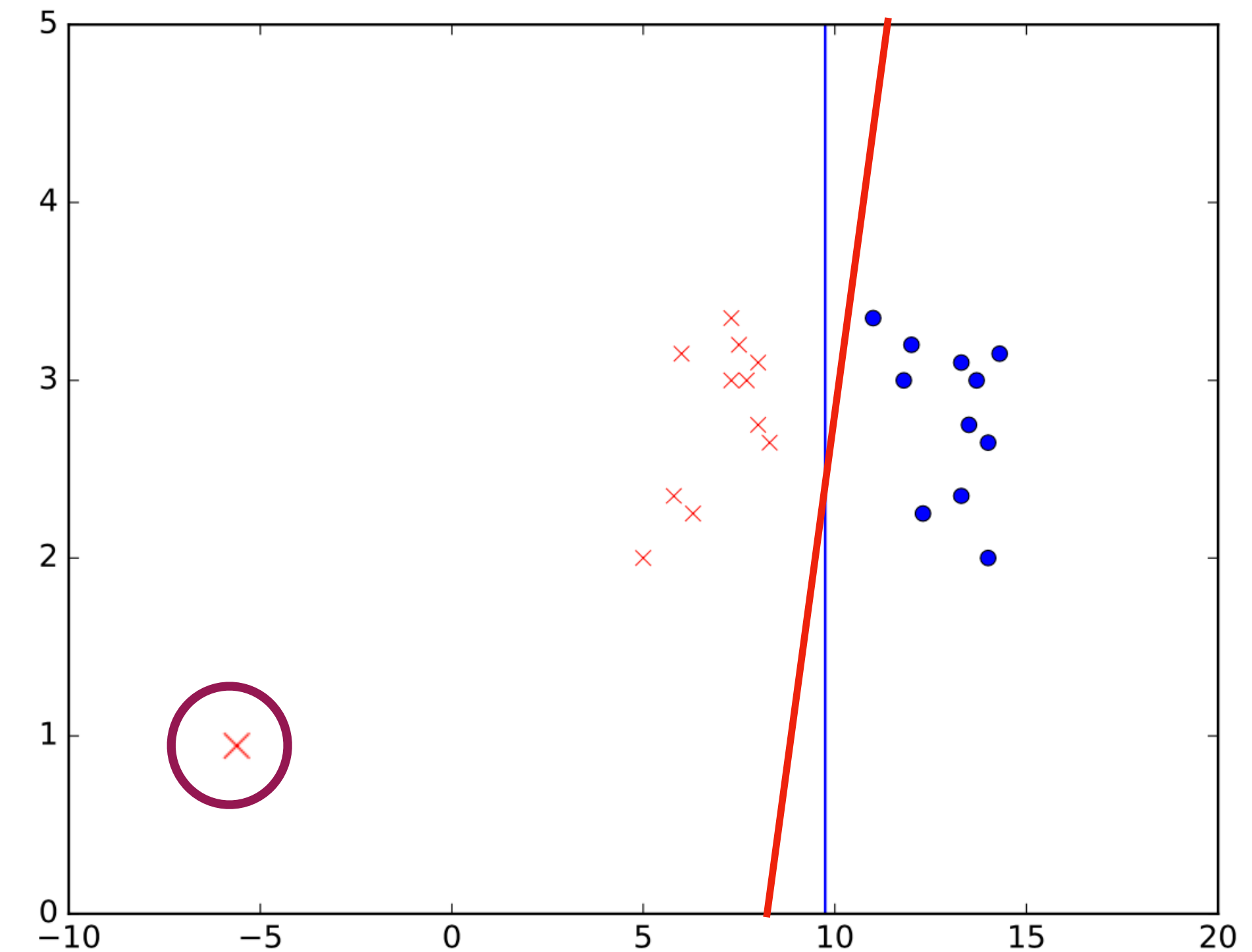
If the error becomes twice as large, the penalty roughly becomes twice as large—not four times as large.

Therefore, the model may decide that it is better to **leave this single outlier misclassified** rather than move the boundary enough to hurt many of the other correctly classified examples.

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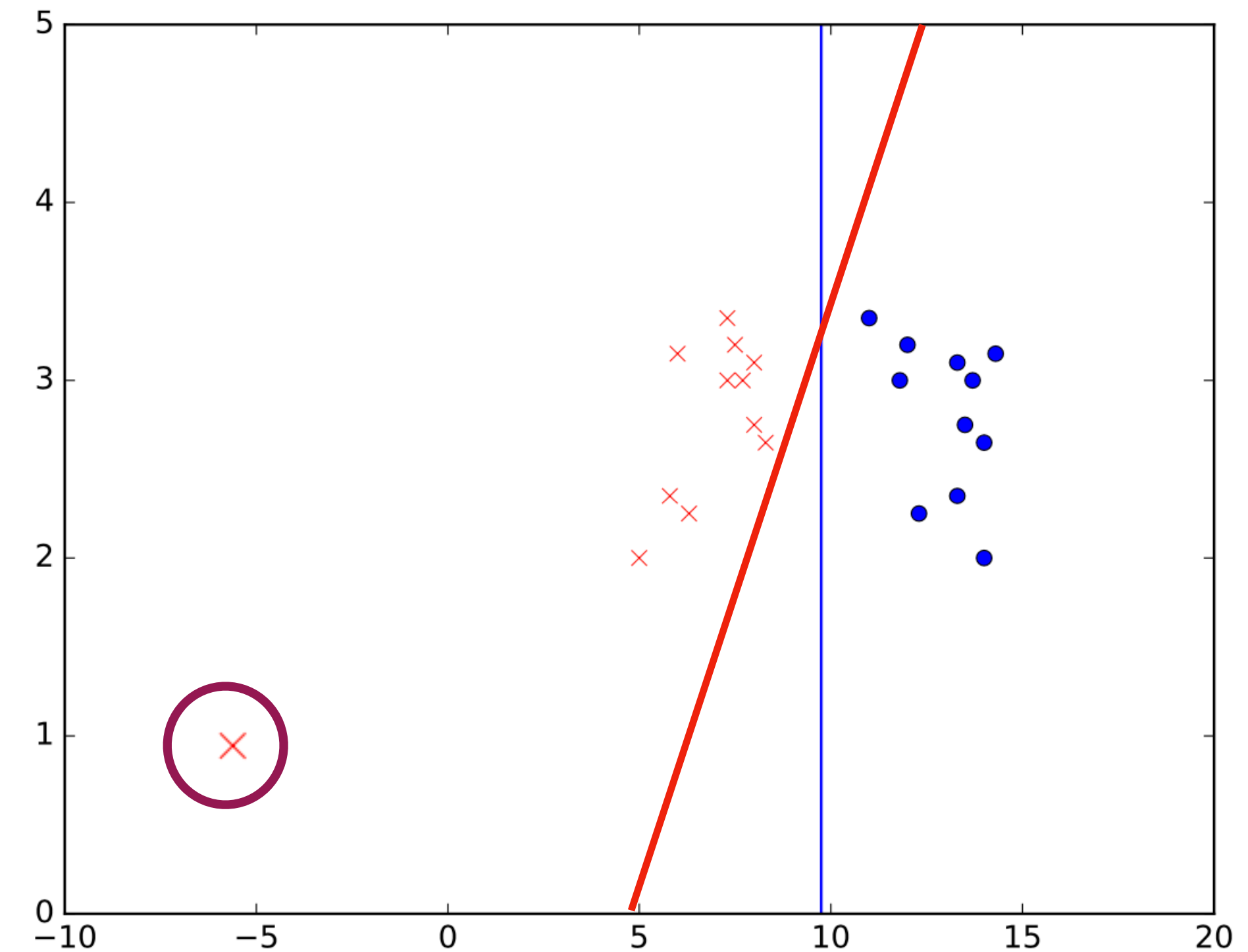
For L_2 : I would draw the new boundary **close to the existing blue vertical line**, somewhere between the main red and blue groups. It may move slightly, but it will likely still separate the two main clusters and sacrifice the new outlier.



Question 4

- Let's assume we want to separate the dataset in this figure using the loss functions

For L_1 : I would draw a boundary that is **more noticeably shifted or tilted toward the new point at (-5,1)**. The exact line cannot be calculated from the figure alone because we don't have the exact coordinates of every point, but the key idea is that it should be **more distorted by the outlier** than the hinge-loss boundary.



Squared loss is more sensitive to outliers than hinge loss.

Question 5

- You are given the choice of two loss functions for a binary classification problem:

the exponential
and logistic losses

$$L(\theta^T x, y) = \exp(-y\theta^T x)$$

$$L(\theta^T x, y) = \log(1 + \exp(-y\theta^T x)).$$

The label $y^{(i)}$ is incorrect for about 10% (the precise number is unimportant) of the training $\{(x^{(i)}, y^{(i)})\}_{i=1}^m$

You will choose a hypothesis by minimizing $J(\theta)$ for one of the two losses.

Which loss is more likely to have better generalization performance? Justify your answer.

$$J(\theta) = \frac{1}{m} \sum_{i=1}^m L(\theta^T x^{(i)}, y^{(i)})$$

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The logistic loss.

It is less sensitive to noise (or mistaken labels), as it grows only linearly for mis-classifications rather than exponentially.

So the exponential loss will work very hard to classify mis-labeled examples correctly.

Question 6

Let X be a matrix of data points in $\mathbb{R}^3$ (each row is a data point, and each column is a different feature). Suppose the second column is exactly twice the first column, and the third column is exactly three times the first.

Find the covariance matrix of X .

Question 6

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X is a 3D matrix.

$$x_2 = 2x_1, \quad x_3 = 3x_1 \qquad X_c = \begin{bmatrix} a & 2a & 3a \end{bmatrix} \qquad a \in \mathbb{R}^n.$$

Covariance matrix formula

$$C = \frac{1}{n-1} X_c^\top X_c$$

$$X_c^\top X_c = \begin{bmatrix} a^\top a & (a^\top)(2a) & (a^\top)(3a) \\ (2a^\top)(a) & (2a^\top)(2a) & (2a^\top)(3a) \\ (3a^\top)(a) & (3a^\top)(2a) & (3a^\top)(3a) \end{bmatrix}$$

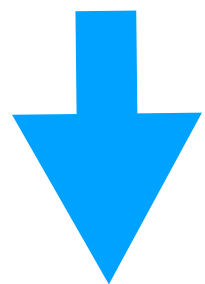
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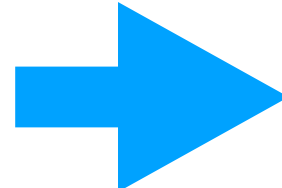
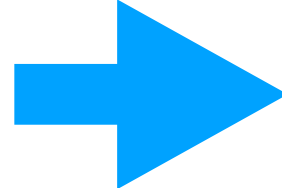
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Simplify (let $\|a\|^2 = a^\top a$):  $X_c^\top X_c = \|a\|^2 \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{bmatrix}$  $C = \frac{\|a\|^2}{n-1} \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{bmatrix}$

Question 7

- A study investigates the relationship between internet usage frequency and preferred device type.

The data is:

Use a chi-square test to examine if internet usage frequency and preferred device type are related.

Usage Frequency	Smartphone	Tablet	Laptop	Desktop
Daily	50	20	60	30
Weekly	30	30	20	20
Monthly	20	30	10	20

Question 7

Compute the row sums, column sums, and grand total

Usage Frequency	Smartphone	Tablet	Laptop	Desktop
Daily	50	20	60	30
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- Daily: $50 + 20 + 60 + 30 = 160$
- Weekly: $30 + 30 + 20 + 20 = 100$
- Monthly: $20 + 30 + 10 + 20 = 80$
- Smartphone: $50 + 30 + 20 = 100$
- Tablet: $20 + 30 + 30 = 80$
- Laptop: $60 + 20 + 10 = 90$
- Desktop: $30 + 20 + 20 = 70$

Usage \ Device	Smartphone	Tablet	Laptop	Desktop	Row total
Daily	50	20	60	30	160
Weekly	30	30	20	20	100
Monthly	20	30	10	20	80
Col total	100	80	90	70	340

H_0 : Null Hypothesis: internet usage frequency and preferred device type are **independent**.

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Under the null hypothesis (usage and device independent), the expected count in cell (i,j) is:

Expected Frequencies

$$E_{ij} = \frac{(\text{row total of } i) \times (\text{column total of } j)}{N}$$

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Daily	47.06	37.65	42.35	32.94
Weekly	29.41	23.53	26.47	20.59
Monthly	23.53	18.82	21.18	16.47
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$$E_{ij} = \frac{(\text{row total of } i) \times (\text{column total of } j)}{N}$$

For example, for **Daily + Smartphone**:

$$E = \frac{160 \times 100}{340} \approx 47.06$$

Perform Chi-Square Test

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

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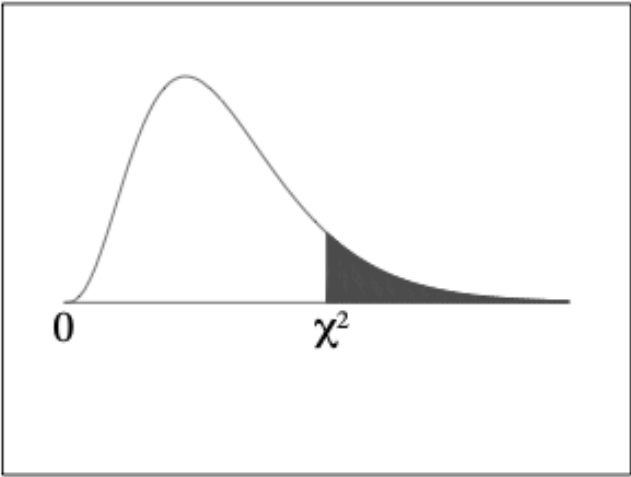
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After summing over all 12 cells:

$\chi^2 \approx 33.28$

Chi-Square Distribution Table



The shaded area is equal to α for $\chi^2 = \chi^2_{\alpha}$.

df	$\chi^2_{.995}$	$\chi^2_{.990}$	$\chi^2_{.975}$	$\chi^2_{.950}$	$\chi^2_{.900}$	$\chi^2_{.100}$	$\chi^2_{.050}$	$\chi^2_{.025}$	$\chi^2_{.010}$	$\chi^2_{.005}$
1	0.000	0.000	0.001	0.004	0.016	2.706	3.841	5.024	6.635	7.879
2	0.010	0.020	0.051	0.103	0.211	4.605	5.991	7.378	9.210	10.597
3	0.072	0.115	0.216	0.352	0.584	6.251	7.815	9.348	11.345	12.838
4	0.207	0.297	0.484	0.711	1.064	7.779	9.488	11.143	13.277	14.860
5	0.412	0.554	0.831	1.145	1.610	9.236	11.070	12.833	15.086	16.750
6	0.676	0.872	1.237	1.635	2.204	10.645	12.592	14.449	16.812	18.548
7	0.989	1.239	1.690	2.167	2.833	12.017	14.067	16.013	18.475	20.278
8	1.344	1.646	2.180	2.733	3.490	13.362	15.507	17.535	20.090	21.955
9	1.735	2.088	2.700	3.325	4.168	14.684	16.919	19.023	21.666	23.589
10	2.156	2.558	3.247	3.940	4.865	15.987	18.307	20.483	23.209	25.188
11	2.603	3.053	3.816	4.575	5.578	17.275	19.675	21.920	24.725	26.757
12	3.074	3.571	4.404	5.226	6.304	18.549	21.026	23.337	26.217	28.300
13	3.565	4.107	5.009	5.892	7.042	19.812	22.362	24.736	27.688	29.819
14	4.075	4.660	5.629	6.571	7.790	21.064	23.685	26.119	29.141	31.319
15	4.601	5.229	6.262	7.261	8.547	22.307	24.996	27.488	30.578	32.801
16	5.142	5.812	6.908	7.962	9.312	23.542	26.296	28.845	32.000	34.267
17	5.697	6.408	7.564	8.672	10.085	24.769	27.587	30.191	33.409	35.718
18	6.265	7.015	8.231	9.390	10.865	25.989	28.869	31.526	34.805	37.156
19	6.844	7.633	8.907	10.117	11.651	27.204	30.144	32.852	36.191	38.582
20	7.434	8.260	9.591	10.851	12.443	28.412	31.410	34.170	37.566	39.997
21	8.034	8.897	10.283	11.591	13.240	29.615	32.671	35.479	38.932	41.401
22	8.643	9.542	10.982	12.338	14.041	30.813	33.924	36.781	40.289	42.796
23	9.260	10.196	11.689	13.091	14.848	32.007	35.172	38.076	41.638	44.181
24	9.886	10.856	12.401	13.848	15.659	33.196	36.415	39.364	42.980	45.559
25	10.520	11.524	13.120	14.611	16.473	34.382	37.652	40.646	44.314	46.928
26	11.160	12.198	13.844	15.379	17.292	35.563	38.885	41.923	45.642	48.290
27	11.808	12.879	14.573	16.151	18.114	36.741	40.113	43.195	46.963	49.645
28	12.461	13.565	15.308	16.928	18.939	37.916	41.337	44.461	48.278	50.993
29	13.121	14.256	16.047	17.708	19.768	39.087	42.557	45.722	49.588	52.336
30	13.787	14.953	16.791	18.493	20.599	40.256	43.773	46.979	50.892	53.672
40	20.707	22.164	24.433	26.509	29.051	51.805	55.758	59.342	63.691	66.766
50	27.991	29.707	32.357	34.764	37.689	63.167	67.505	71.420	76.154	79.490
60	35.534	37.485	40.482	43.188	46.459	74.397	79.082	83.298	88.379	91.952
70	43.275	45.442	48.758	51.739	55.329	85.527	90.531	95.023	100.425	104.215
80	51.172	53.540	57.153	60.391	64.278	96.578	101.879	106.629	112.329	116.321
90	59.196	61.754	65.647	69.126	73.291	107.565	113.145	118.136	124.116	128.299
100	67.328	70.065	74.222	77.929	82.358	118.498	124.342	129.561	135.807	140.169

Question 7

Perform Chi-Square Test

Usage	Smartphone	Tablet	Laptop	Desktop
Daily	47.06	37.65	42.35	32.94
Weekly	29.41	23.53	26.47	20.59
Monthly	23.53	18.82	21.18	16.47

Total $\chi^2 \approx 33.28$

Degrees of freedom: $(3 - 1) \times (4 - 1) = 6$

At $\alpha = 0.05$, critical $\chi^2(6) \approx 12.59$; at $\alpha = 0.01$, ≈ 16.81 .

Since **33.28 > 16.81**, we **reject the null of independence** (even at 1%).

Small p-value $\rightarrow$ the observed data would be unusual under $H_0 \rightarrow$ evidence **against** H_0 .

Large p-value $\rightarrow$ the observed data are reasonably compatible with $H_0 \rightarrow$ **not enough evidence to reject** H_0 .

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Conclusion

Internet usage frequency and preferred device type are **significantly associated**.

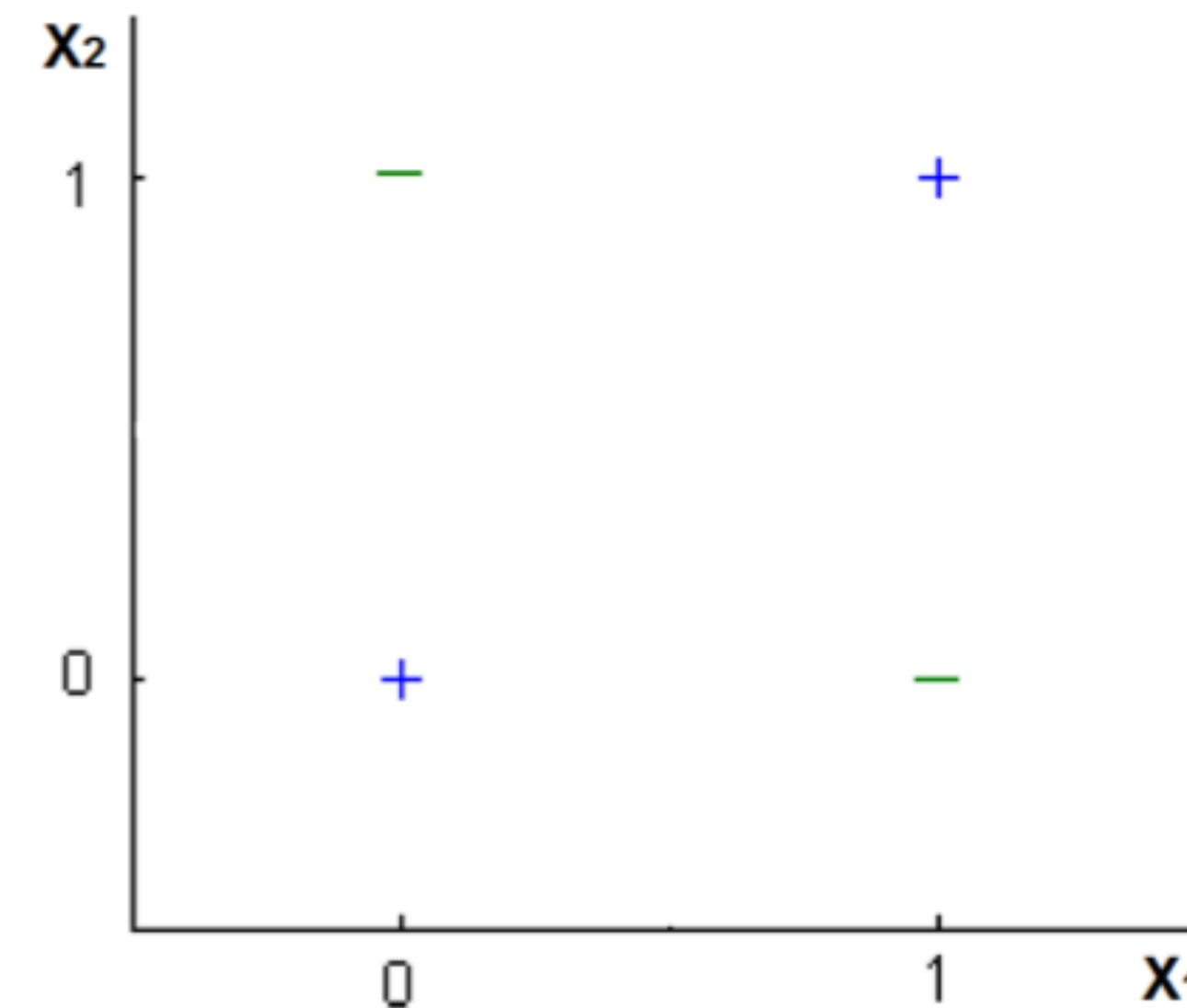
Question 8

- Suppose we learn a Naive Bayes classifier from the examples in Figure 1, using MLE (maximum likelihood estimation) the training rule.

Write down all the parameters and their estimated values (note: both $P(Y)$ and $P(X_i|Y)$ should be Bernoulli distributions).

Also, does this learned Naive Bayes perfectly classify the four examples?

Figure 1: Two positive examples and two negative examples.

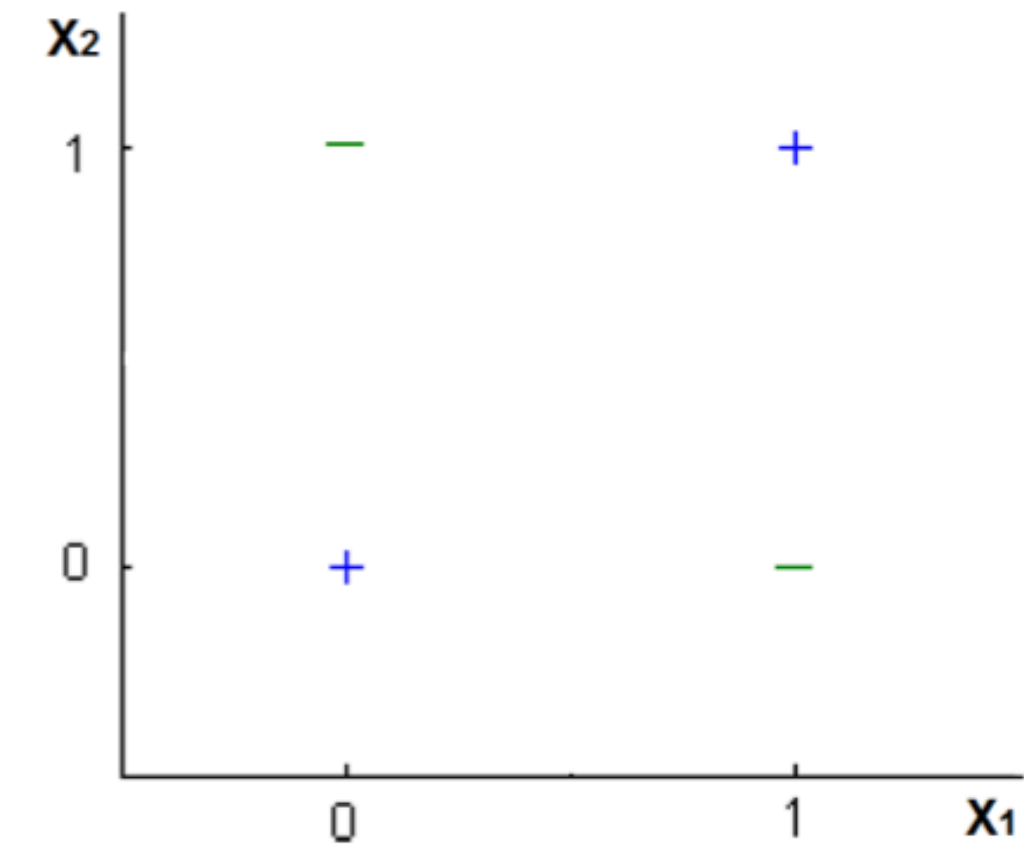


Question 8

We have a **Naive Bayes classifier** with:

- Two binary features: $X_1, X_2 \in \{0, 1\}$.
- One binary label: $Y \in \{0, 1\}$.

Figure 1: Two positive examples and two negative examples.



MLE estimate for class prior is just relative frequency:

Prior probabilities

$$P(Y = 1) = \frac{\#\{Y = 1\}}{\text{total}}, \quad P(Y = 0) = \frac{\#\{Y = 0\}}{\text{total}}$$

Conditional probabilities

For each feature X_i given label Y :

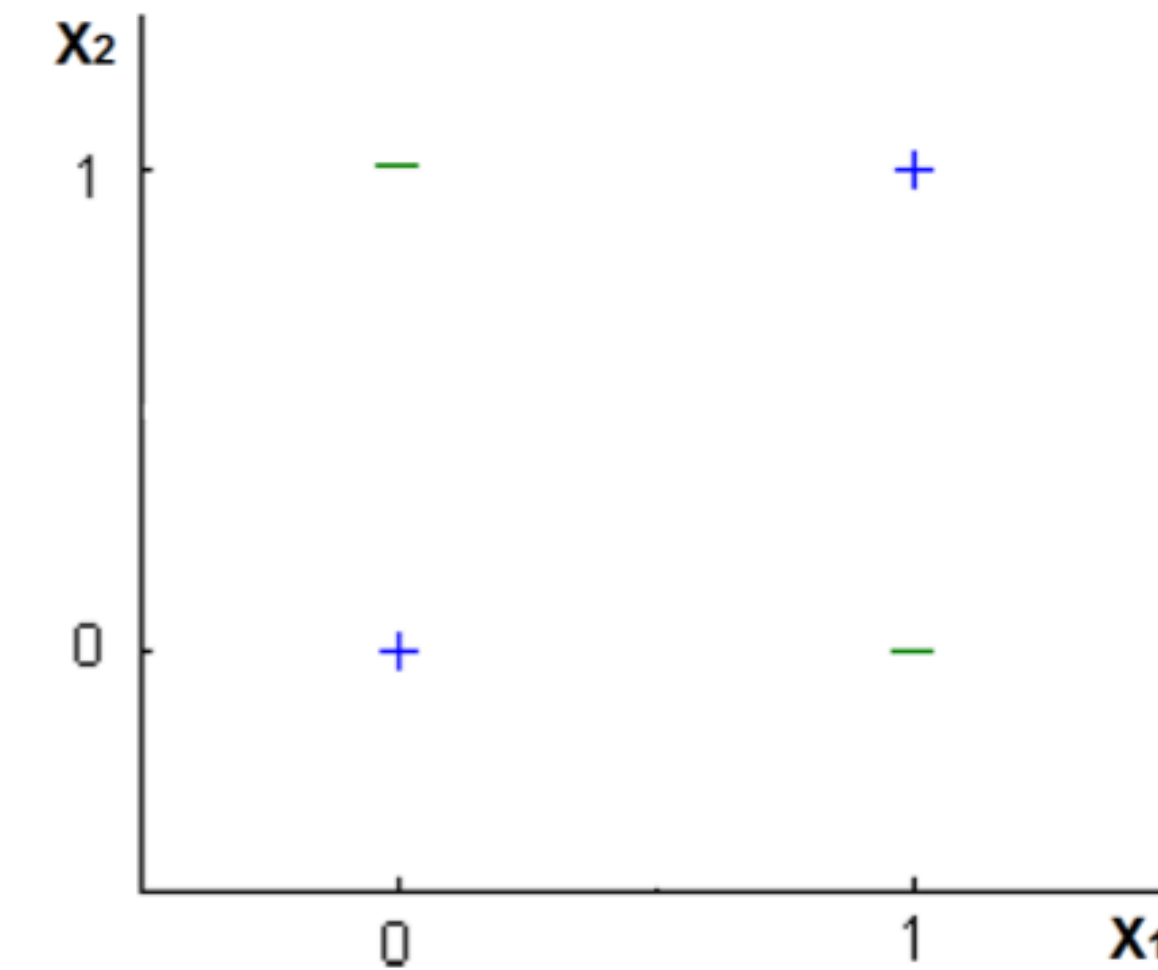
$$P(X_i = 1 \mid Y = y) = \frac{\#\{\text{examples with } X_i = 1, Y = y\}}{\#\{Y = y\}}$$

Question 8

Figure 1: Two positive examples and two negative examples.

We have a **Naive Bayes classifier** with:

- Two binary features: $X_1, X_2 \in \{0, 1\}$.
- One binary label: $Y \in \{0, 1\}$.



$$P(Y = 1) = 0.5$$

$$P(Y = 0) = 0.5$$

$$P(X_1 = 1|Y = 0) = P(X_1 = 1|Y = 1) = 0.5$$

$$P(X_1 = 0|Y = 0) = P(X_1 = 0|Y = 1) = 0.5$$

$$P(X_2 = 1|Y = 0) = P(X_2 = 1|Y = 1) = 0.5$$

$$P(X_2 = 0|Y = 0) = P(X_2 = 0|Y = 1) = 0.5$$

So all conditional probabilities are 0.5

This is a very poor classifier since for any X_1, X_2 it will predict $P(Y = 1 | X_1, X_2) = P(Y = 0 | X_1, X_2) = 0.5$.

Naturally, it cannot perfectly classify the examples in the figure.

Question 9

Consider a decision tree used for binary classification. At a certain node, there are 100 samples, with 60 belonging to class A and 40 belonging to class B.

- (a) Compute the misclassification error, Gini index, and cross-entropy (entropy) at this node.
- (b) Which measure better captures gradual changes in node purity for split selection(e.g., going from 60/40 to 65/35)? Explain briefly.

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$$p_A = \frac{60}{100} = 0.6, \quad p_B = \frac{40}{100} = 0.4.$$

1. Misclassification error

$$L_{\text{misclass}} = 1 - \max(p_A, p_B) = 1 - 0.6 = 0.40$$

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3. Cross-entropy $H = - \sum_c p_c \log_2 p_c$

$$H = - [0.6 \log_2(0.6) + 0.4 \log_2(0.4)] = 0.971$$

Question 9

Consider a decision tree used for binary classification. At a certain node, there are 100 samples, with 60 belonging to class A and 40 belonging to class B.

(b) Which measure better captures gradual changes in node purity for split selection(e.g., going from 60/40 to 65/35)? Explain briefly.

In the usual **decision-tree interpretation, entropy and Gini are more useful/sensitive for evaluating candidate splits** because they change smoothly with class purity. Misclassification error is comparatively coarse and often fails to distinguish well between candidate splits that have the same majority class.

For example, changing from **60/40 to 65/35** gives:

Gini: 0.48 -> 0.455

Entropy: 0.971-> 0.934

Both decrease as the node becomes purer.

Entropy (and similarly Gini) is more sensitive to changes in purity than misclassification error, which is why entropy or Gini is generally preferred when choosing decision-tree splits.

Question 10

We use a 1D linear regression model: $\hat{y} = wx + b$

dataset: $D = \{(1,2), (2,3), (3,5)\}$

We use MSE for loss and Batch Gradient Descent to update weights.

- (a) Compute predictions and errors as $w=0$ and $b=0$
- (b) Write the gradient for batch GD w.r.t w .
- (c) Compute gradient of loss with respect to w and gradient of loss with respect to b at $w=0$ and $b=0$.



Questions?

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